Transfer: energy and work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose and justify energy methods in new physical situations.

SUCCESS CRITERIA - I can...

1. Choose the correct energy terms and units.
2. Show a consistent numerical balance.
3. Explain model assumptions and limitations.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “work” and “power” to explain your model or result. Sentence starters are optional.

work: energy transferred by a force along a displacement

power: rate of energy transfer

energy budget: an account of all energy stored and transferred

01 Warm-up: bring back an earlier idea

At fixed mass, speed becomes 1/3 of its original value. **Compare** kinetic energy and the energy-equivalent TNT amount.

Word help: kinetic energy = energy associated with motion

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $W_{net} = \Delta K$; $K = \frac{1}{2}mv^2$; $U_g = mgh$; $U_s = \frac{1}{2}kx^2$
$P = W/\Delta t$.
$g = 10$ N/kg where needed. Select relationships for the stated system and conditions.

Worked reasoning: A 2 kg cart drops 2 m with g =10, then loses 10 J to friction. From rest, its final kinetic energy is 40-10=30 J, so speed $\sqrt{30} \approx 5.48$ m/s.

Your turn: Draw a complete energy account and a sketch identifying where friction acts.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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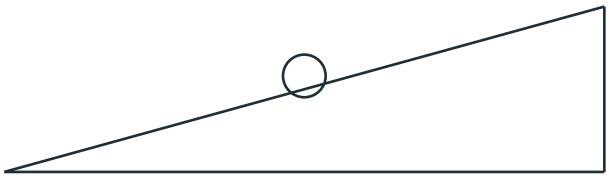
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A cart coasts down a frictionless ramp from rest.
Explain which energy decreases and which increases.
Mark the vertical height in the sketch.

Sentence starter: My claim is … . The diagram/data show …, so … .



The marker represents the object on the sloping surface.

Marzano check - circle: 1 2 3

04 Apply the physics

A 2 kg cart drops 2 m.
Use g=10.
Find initial gravitational energy and bottom kinetic energy if losses are negligible.
Use the energy model to show the quantities you compare; label units and any unknowns.

Word help: negligible = small enough to leave out of this model

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A motor supplies 100 J while the load gains 70 J mechanical energy.
Explain where the other 30 J might go and whether energy was destroyed.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An ideal system has total mechanical energy 24 J. At a point where U=4 J, find K.
Show the energy changes and transfers; label units.

Word help: mechanical energy = total kinetic and potential energy in a model

Sentence starter: I found … [unit]. This value means … in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

A device transfers 60 J in 5 s.
Find its average power. It then transfers the same 60 J in 6 s. Complete the comparison table and describe what changes.

Sentence starter: I found … [unit]. This value means … in this situation.

Run	Energy (J)	Time (s)	Power (W)
A	60	5	____
B	60	6	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. Change one feature so the same cart arrives with 20 J kinetic energy.
Show two different valid designs.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3